

Stationary models II

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Prediction-error decomposition of likelihood function

- Using the definition of conditional densities, we can decompose the likelihood function of *any* vector time series $\{y_t\}$ as

$$p(y_1, \dots, y_T \mid \theta) = p(y_1 \mid \theta) \prod_{t=2}^T p(y_t \mid \{y_\tau\}_{1 \leq \tau \leq t-1}, \theta).$$

- Called the **prediction-error decomposition** (or factorization), since the terms in the product are one-step-ahead predictive densities.
- Note: Does not require stationarity.

Gaussian likelihood function

- If $\{y_t\}$ is an n -dim. Gaussian time series, then

$$\log p(y_1, \dots, y_T \mid \theta) = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \log \det(\mathbf{\Gamma}(\theta)) \\ - \frac{1}{2}(\mathbf{Y} - \boldsymbol{\mu}(\theta))' \mathbf{\Gamma}(\theta)^{-1} (\mathbf{Y} - \boldsymbol{\mu}(\theta)),$$

where $\mathbf{Y} \equiv (y_1', \dots, y_T')'$, and $\boldsymbol{\mu}(\theta)$ and $\mathbf{\Gamma}(\theta)$ are the mean and var-cov matrix of $\mathbf{Y}$ implied by μ_y and $\Gamma_y(\cdot)$, given parameters θ .

- Define

$$\hat{e}_t(\theta) \equiv y_t - E(y_t \mid \{y_\tau\}_{1 \leq \tau \leq t-1}, \theta), \quad \hat{V}_t(\theta) \equiv \text{Var}(\hat{e}_t \mid \theta).$$

Then the prediction-error decomposition is

$$\log p(y_1, \dots, y_T \mid \theta) = -\frac{nT}{2} \log(2\pi) \\ - \frac{1}{2} \sum_{t=1}^T \left\{ \log \det(\hat{V}_t(\theta)) + \hat{e}_t(\theta)' \hat{V}_t(\theta)^{-1} \hat{e}_t(\theta) \right\}.$$

VAR inference: Gaussian likelihood function

- The Gaussian likelihood is particularly tractable for Gaussian VARs.
- Letting $\theta \equiv (\nu, A_1, \dots, A_p, \Sigma)$, we get, for $t \geq p + 1$,

$$\hat{e}_t(\theta) = y_t - \nu - \sum_{\ell=1}^p A_\ell y_{t-\ell}, \quad \hat{V}_t(\theta) = \Sigma.$$

- For $t = 1, \dots, p$, the formula is not so simple. But can easily compute

$$p(y_1, \dots, y_p \mid \theta)$$

using the mean and ACF implied by the VAR. Then

$$p(y_1, \dots, y_T \mid \theta) = p(y_1, \dots, y_p \mid \theta) p(y_{p+1}, \dots, y_T \mid y_1, \dots, y_p, \theta),$$

with the last factor computed using the prediction-error decomp.

- **Exercise:** Derive the VAR likelihood for the case where the innovations u_t are non-Gaussian.

VAR inference: least-squares estimator

- In practice, researchers often use only the *conditional* likelihood

$$p(y_{p+1}, \dots, y_T \mid y_1, \dots, y_p, \theta).$$

- Corresponds to the likelihood in the multivariate linear regr. model

$$y_t = \beta x_t + u_t, \quad u_t \stackrel{iid}{\sim} N(0, \Sigma),$$

where $k = 1 + np$ and

$$\beta \equiv (\nu, A_1, \dots, A_p) \in \mathbb{R}^{n \times k}, \quad x_t \equiv (1, y'_{t-1}, \dots, y'_{t-p})' \in \mathbb{R}^k.$$

(For the likelihood, it doesn't matter that x_t is not i.i.d.)

- The MLE is the least-squares solution:

$$\begin{aligned} \hat{\beta} &= (\sum_{t=p+1}^T y_t x_t') (\sum_{t=p+1}^T x_t x_t')^{-1}, \\ \hat{\Sigma} &= \frac{1}{T-p} \sum_{t=p+1}^T \underbrace{(y_t - \hat{\beta} x_t)(y_t - \hat{\beta} x_t)'}_{\text{residuals } \hat{u}_t}. \end{aligned}$$

VAR inference: stationary asymptotics

- Just as in the cross-sectional multiv. linear regression model,

$$\begin{aligned}\hat{\beta} - \beta &= (\sum_t u_t x_t') (\sum_t x_t x_t')^{-1}, \\ \implies \text{vec}(\hat{\beta} - \beta) &= (\hat{S}_x^{-1} \otimes I_n) \text{vec}(\frac{1}{T-p} \sum_t x_t \otimes u_t), \\ \text{where } \hat{S}_x &\equiv \frac{1}{T-p} \sum_t x_t x_t' .\end{aligned}$$

- $(x_t', u_t)'$ is not i.i.d., so we can't apply the usual LLN and CLT.
- But as long as $x_t \otimes u_t$ is a *martingale difference sequence* (and some moments exist), we'll see later that the same results apply. Then

$$\begin{aligned}\sqrt{T-p} \text{vec}(\hat{\beta} - \beta) &\xrightarrow{d} N(0, \Omega) \quad \text{as } T \rightarrow \infty, \\ \Omega &\equiv (S_x^{-1} \otimes I_n) \text{Var}(x_t \otimes u_t) (S_x^{-1} \otimes I_n), \quad S_x \equiv E(x_t x_t').\end{aligned}$$

- Exercise:** Show that if the VAR(p) model is stationary with i.i.d. innovations u_t , then $x_t \otimes u_t$ is a MDS, and $\Omega = S_x^{-1} \otimes \Sigma$.

VAR inference: hypothesis testing

- Under weak conditions, we have the consistent estimator

$$\hat{\Omega} \equiv (\hat{S}_x^{-1} \otimes I_n) \left(\frac{1}{T-p} \sum_t (x_t x_t' \otimes \hat{u}_t \hat{u}_t') \right) (\hat{S}_x^{-1} \otimes I_n) \xrightarrow{p} \Omega.$$

Or if we assume i.i.d. innovations u_t :

$$\hat{\Omega} \equiv (\hat{S}_x^{-1} \otimes \hat{\Sigma}) \xrightarrow{p} \Omega.$$

- Can test any linear null hypothesis $H_0: R \text{vec}(\beta) = b$ against the two-sided alternative $H_1: R \text{vec}(\beta) \neq b$ using the Wald statistic

$$\hat{W} = (T-p)(R \text{vec}(\hat{\beta}) - b)' (R \hat{\Omega} R')^{-1} (R \text{vec}(\hat{\beta}) - b)$$

with asy. distribution χ_r^2 under H_0 , if R is $r \times nk$ with full rank.

Application: Why do stock prices react to the Fed?

- Bernanke & Kuttner (2005) isolate changes in the prices of futures written on the Federal Funds Rate, in a 1-day window around Federal Open Market Committee announcements.
- These *monetary policy shocks* are correlated with stock returns.
- Is the market excess return e_t^x driven by news about future log dividends d_t , real interest rates r_t , or returns x_t ?
- Campbell & Shiller (1988) decomposition:

$$\begin{aligned} e_{t+1}^x &= (E_{t+1} - E_t) \sum_{\ell=0}^{\infty} \rho^{\ell} \Delta d_{t+1+\ell} \\ &\quad - (E_{t+1} - E_t) \sum_{\ell=0}^{\infty} \rho^{\ell} r_{t+1+\ell} \\ &\quad - (E_{t+1} - E_t) \sum_{\ell=1}^{\infty} \rho^{\ell} x_{t+1+\ell}. \end{aligned}$$

Application: Why do stock prices react to the Fed? (cont.)

- Assume $y_t \equiv (x_t, r_t, w_t')'$ follows a VAR(1):

$$y_t = \nu + Ay_{t-1} + u_t, \quad u_t \stackrel{iid}{\sim} WN(0, \Sigma).$$

Here w_t are additional var's. Could generalize to VAR(p) via companion form.

- VAR implies

$$(E_{t+1} - E_t) \sum_{\ell=0}^{\infty} \rho^{\ell} r_{t+1+\ell} = s_r' (I - \rho A)^{-1} u_{t+1},$$

where s_r is a vector s.t. $r_t = s_r' y_t$. **Exercise:** Verify above formula.

- Same for forecasts of future x_t . Obtain responses of Δd_t as residual.
- Get response to monetary policy by regressing $\hat{u}_t$ on Fed Funds shock.
- Calculate standard errors using delta method, since everything is a function of the VAR parameters (ρ calibrated).

Application: Why do stock prices react to the Fed? (cont.)

The Impact of Monetary Policy on Dividends, Interest Rates, and Future Returns

The table reports the impact of monetary policy surprises on the current excess equity return and the discounted sums of future excess equity returns, current and future real interest rates, and current and future dividends. The six-variable VAR(1) used to construct real interest rate and excess equity return forecasts is estimated over the sample indicated in the column headings, and the contemporaneous response to the funds rate surprises is estimated on the February 1994–December 2002 subsample. Parentheses contain t -statistics, calculated using the delta method.

	Sample Used for VAR	
	1/73–12/02	5/89–12/02
Current excess return	–11.55 (3.87)	–11.01 (3.72)
Future excess returns	6.10 (1.74)	3.29 (1.10)
Real interest rate	0.64 (1.03)	0.77 (1.87)
Dividends	–4.82 (1.73)	–6.96 (2.35)

VAR inference: small-sample bias

- Since the (conditional) Gaussian VAR likelihood looks exactly like the i.i.d. Gaussian regression likelihood, it's tempting to think that the VAR estimator is unbiased in finite samples, like cross-sectional OLS. But this is wrong!
- Unlike in cross-sectional regression, in the VAR model the errors u_t are not **strictly exogenous**: $E[u_t \mid x_1, x_2, \dots, x_T] \neq 0$. This is the case even if the innovations u_t are i.i.d.
- The problem is caused by lagged endogenous regressors $x_t = (1, y'_{t-1}, \dots, y'_{t-p})'$:

$$E[u_t \mid x_{t+1}] = E[u_t \mid y_t, y_{t-1}, \dots, y_{t-p+1}] \neq 0,$$

since y_t depends on u_t according to the VAR model.

- The VAR coefficient estimator is biased in finite samples (though consistent as $T \rightarrow \infty$).

VAR inference: small-sample bias (cont.)

- Suppose we apply the usual VAR(1) least-squares estimator, but the true model is such that each var. y_{it} follows the same AR(1):

$$y_{it} = \rho y_{i,t-1} + u_{it}, \quad u_{it} \stackrel{iid}{\sim} WN(0, \sigma^2), \quad i = 1, \dots, n,$$

with u_{it} uncorrelated across i .

- Then the bias of the least-squares estimator (with intercept) is

$$E(\hat{A}_1) - A_1 = -\frac{1 + (2 + n)\rho}{T} I_n + O(T^{-3/2}).$$

- Diagonal of $\hat{A}_1$ downward biased for small T . Higher persistence or $n \Rightarrow$ larger absolute bias. Also typically true for more general DGPs.
- See Kilian & Lütkepohl (2017), chapter 2.3.3, for general formula.
- Could correct this bias using the formula (might increase variance).

Bayesian VARs

- Until now, considered frequentist inference: care about performance of procedures across *repeated hypothetical histories of the world*.
- Sometimes it makes more sense to adopt a Bayesian perspective: What is optimal given the specific *actual* history of the world?
- Through the prior distribution, Bayesian inference also makes it convenient to impose **shrinkage** to deal with small samples and high-dimensional systems.
- For stationary, low-dimensional time series models, Bayesian and frequentist inference often similar (the Bernstein-von Mises Theorem). Not true more generally.

Bayesian VARs: normal prior, fixed Σ

- For simplicity, suppose that $\Sigma = \text{Var}(u_t)$ is known. In practice, can set $\Sigma = \hat{\Sigma}$ (least-squares estimate).
- Suppose we have a normal prior

$$\text{vec}(\beta) \sim N(\beta^*, V_\beta).$$

- Using the conditional Gaussian VAR likelihood, posterior density is

$$\begin{aligned} p(\beta \mid y_1, \dots, y_T) &\propto p(y_{p+1}, \dots, y_T \mid y_1, \dots, y_p, \beta) \pi(\beta) \\ &\propto \exp \left\{ -\frac{1}{2} (\text{vec}(\beta) - \bar{\beta})' \bar{V}_\beta^{-1} (\text{vec}(\beta) - \bar{\beta}) \right\}, \end{aligned}$$

$$\bar{\beta} \equiv \bar{V}_\beta \left((\mathbf{X} \otimes \Sigma^{-1}) \text{vec}(\mathbf{Y}) + V_\beta^{-1} \beta^* \right), \quad \bar{V}_\beta \equiv \left((\mathbf{X}\mathbf{X}' \otimes \Sigma^{-1}) + V_\beta^{-1} \right)^{-1},$$

$$\mathbf{X} \equiv (x_{p+1}, \dots, x_T) \in \mathbb{R}^{k \times (T-p)}, \quad \mathbf{Y} \equiv (y_{p+1}, \dots, y_T) \in \mathbb{R}^{n \times (T-p)}.$$

- See Kilian & Lütkepohl (2017), ch. 5.2.2, for derivation.

Bayesian VARs: normal prior, fixed Σ (cont.)

- Hence, the posterior distribution of the parameter vector $\text{vec}(\beta)$ is normal with mean $\bar{\beta}$ and variance $\bar{V}_\beta$.
- As the prior becomes infinitely diffuse ($V_\beta^{-1} \rightarrow 0_{nk \times nk}$), we have

$$\bar{\beta} \rightarrow \text{vec}(\hat{\beta}), \quad \bar{V}_\beta \rightarrow \frac{1}{T-p}(\hat{S}_x^{-1} \otimes \Sigma),$$

the least-squares estimator and frequentist sampling variance.

- But in small samples, can select the prior mean/variance to **shrink** the unrestricted least-squares estimate toward *a priori* reasonable values, to reduce overfitting. Feasible even when $\dim(\beta) > T - p$.

Bayesian VARs: Minnesota prior

- How to pick prior mean β^* and variance V_β ?
- Popular choice: **Minnesota prior** (Doan, Litterman & Sims, 1984). Choose β^* and *diagonal* V_β such that

$$E(A_{ii,1}) = 1, \quad E(A_{ij,\ell}) = 0 \text{ otherwise,}$$

$$\text{Var}(A_{ij,\ell}) = \begin{cases} (\lambda/\ell)^2 & \text{if } i = j, \\ (\lambda\xi/\ell)^2 \frac{\Sigma_{ii}}{\Sigma_{jj}} & \text{if } i \neq j, \end{cases}$$

$$\text{Var}(\nu_i) = \infty.$$

- Shrinks toward independent random walks. Overall degree of shrinkage governed by λ .
- Smaller prior variance on coef's on larger lags. With $\xi < 1$, smaller prior variance on coef's on “non-own” lags $A_{ij,\ell}$, $i \neq j$.
- Litterman (1986): $\xi = 0.3$, $\lambda = 0.2$ works well for U.S. macro data.

Bayesian VARs: normal-inverse-Wishart prior

- To allow for *a priori* randomness in Σ , we can adopt an n -dimensional **inverse-Wishart** prior

$$\Sigma \sim IW(\Psi, d).$$

Then $E(\Sigma) = \frac{1}{d-n-1}\Psi$. The density function is

$$\pi(\Sigma) \propto \det(\Sigma)^{-\frac{d+n+1}{2}} e^{-\frac{1}{2} \text{tr}(\Psi \Sigma^{-1})}.$$

For additional info on the IW distribution, see Wikipedia.

- Turns out to be tractable to impose prior

$$(\text{vec}(\beta') \mid \Sigma) \sim N(b^*, \Sigma \otimes \Omega),$$

where $\Omega \in \mathbb{R}^{k \times k}$ is positive definite and chosen by the researcher.

- This prior $\pi(\beta \mid \Sigma)\pi(\Sigma)$ is called a **normal-inverse-Wishart** prior.

Bayesian VARs: normal-inverse-Wishart prior (cont.)

- Using the conditional Gaussian VAR likelihood, one can show that the posterior distribution is also of the normal-inverse-Wishart form:

$$(\Sigma \mid \text{data}) \sim IW(\bar{\Psi}, \bar{d}), \quad (\text{vec}(\beta') \mid \Sigma, \text{data}) \sim N(\bar{b}, \Sigma \otimes \bar{\Omega}).$$

See Giannone, Lenza & Primiceri (2015) for formulas and derivations.

- Easy to simulate from the posterior distribution.

Bayesian VARs: marginal likelihood

- Suppose we're deciding between two models M_1 and M_2 , e.g., a VAR(1) and VAR(2). The **posterior odds ratio** is given by

$$\frac{p(M_1 | y_1, \dots, y_T)}{p(M_2 | y_1, \dots, y_T)} = \underbrace{\frac{p(y_1, \dots, y_T | M_1)}{p(y_1, \dots, y_T | M_2)}}_{\text{Bayes factor}} \underbrace{\frac{\pi(M_1)}{\pi(M_2)}}_{\text{prior odds ratio}},$$

where for any model M , we define the **marginal likelihood**

$$p(y_1, \dots, y_T | M) = \int p(y_1, \dots, y_T | \theta_M, M) \pi(\theta_M | M) d\theta_M.$$

Here θ_M are the model's param's and $\pi(\theta_M | M)$ the model's prior.

- If all models are *a priori* equally likely, it makes sense to use the model with highest marginal likelihood.
- Marginal likelihood can be computed in closed form for Gaussian VARs with normal or normal-IW priors, see G, L & P (2015).

Bayesian VARs: hierarchical priors

- So far, have treated **prior hyperparameters** (e.g., β^* , V_β) as given.
- There are many “default” suggestions in the literature, e.g., Minnesota prior. See also review in Kilian & Lütkepohl (2017), ch. 5.
- If there are only a few prior hyperparameters, we could select them by maximizing the marginal likelihood (e.g., compute the marg. lik. for a grid of hyperparameter values).
- But we could also adopt a **hierarchical Bayes** approach and endow the prior hyperparameters with their own prior distributions. This way we learn about the hyperparameters from the data.
- For example, could learn from the data about appropriate degree of shrinkage λ in Minnesota prior.
- Requires Markov Chain Monte Carlo methods to simulate from hierarchical posterior distribution. See G, L & P (2015).

Application: Forecasting with large BVARs

- Giannone, Lenza & Primiceri (2015) employ a BVAR with a Minnesota-style prior that shrinks toward nonstationary dynamics with possible cointegration. Hierarchical specification. Fix $p = 5$.
- Forecast using small system (3 var's), medium system (7 var's), and large system (22 var's, yielding $1 + np = 111$ parameters per eq'n!).
- 1959q1–2008q4. Recursive out-of-sample forecasting.

TABLE 2.—MSFE OF POINT FORECASTS

Horizons	Variables	Small (S)		Medium (M)		Large (L)		Factor M	RW
		VAR	BVAR	VAR	BVAR	VAR	BVAR		
One quarter	Real GDP	13.49	9.61	19.15	7.97		8.18	7.29	10.23
	GDP deflator	1.53	1.32	2.26	1.35		1.10	1.14	5.19
	Federal funds rates	1.61	1.04	1.82	1.03		1.00	1.25	1.06
One year	Real GDP	5.40	3.85	12.10	3.42		3.97	3.52	3.98
	GDP deflator	1.61	1.45	2.25	1.58		0.96	1.01	4.65
	Federal funds rates	0.58	0.32	0.56	0.31		0.36	0.32	0.31

The table reports the mean squared forecast errors of the BVARs and the competing models (VAR: flat-prior VAR, RW: random walk in levels with drift; factor M: factor augmented regression), for each variable and horizon. The evaluation sample is 1975Q1–2008Q4 for the one-quarter-ahead forecasts and 1975Q4–2008Q4 for the one-year-ahead forecasts.